MOTIVATION

Bridging model for programming without have to worry about pysical implementation and hardware design to not worry about software.

Introduction of BSP solved the problem → Bulk - synchronous parallel (BSP)

→ 3 Fundamental characteristic of BSP

1→ n° of independent units

2→ Communication network for delivering point-to-point messages between the various components

3→ Barrier synchronization: Hardware or software mechanism to synchronize all components at regular and programmable time intervals (PARAMETER L)

The execution occurs through cycles called supersteps during which are performed operation on local data and receive previous outputs.

PARALLEL SLACKNESS

The programmer is encouraged to program for a number of virtual processor (V) significantly higher than the real number p of processor such as $V = p\log(p)$

Memory management and concurrency

Data distribution problem to avoid bottlenecks:

- Hashing for automatic allocation: distribute the data with pseudo-random hashing function to map symbolic memory addresses into physical addresses
  ↳ With adequate L and sufficient slackness the collision are minimal.

- Simultaneous accesses (CREW PRAM): even cases where multiple threads attempt to read and write to the same variable at the same instant can be simulated on BSP.
  ↳ Data are replicated from source node and rely on sorting algorithm or prob. functions